CAPSTONE / PROJECT REPORT

PC Parts Check

A source-grounded compatibility assistant for CPU, motherboard and RAM.

Catalog scope	Evaluation	Proxy model
4 boards / 7 CPUs / 2 kits	30 labelled prompts	gemma-4-E4B-it

Problem and audience

A first-time PC builder or student upgrading a computer often starts with a list of parts from separate sources. Similar motherboard names can refer to different products, and a CPU's socket alone does not establish board-specific firmware support. The app combines these checks and shows the manufacturer evidence a user can verify.

User workflow

A user enters a Thai or English parts list, optionally with the installed BIOS version. The assistant searches the catalog and calls a rule checker for the named parts. The UI shows the verdict, individual rules, and the sources used. The direct form accepts catalog selections and runs the same rule checker without a model connection.

Project scope

The catalog covers selected ASUS TUF AM4 and AM5 boards, AMD CPUs, and two Kingston memory kits. It contains 12 curated CPU-support rows. The checks address socket, documented CPU support, minimum BIOS, and DDR generation. This keeps the evaluation small enough to label manually and lets the demo trace a conclusion to a source.

Demonstration scenario

The ASUS TUF GAMING B550-PLUS support table lists BIOS 2803 for Ryzen 7 5700X3D. A user can enter BIOS 2423 to see a conditional result, then enter 2803 to see the firmware condition pass. The example tests a board-specific rule on the same hardware.

ASUS TUF GAMING B550-PLUS CPU support

Architecture

Retrieval finds manufacturer evidence. A separate rule checker produces the verdict.

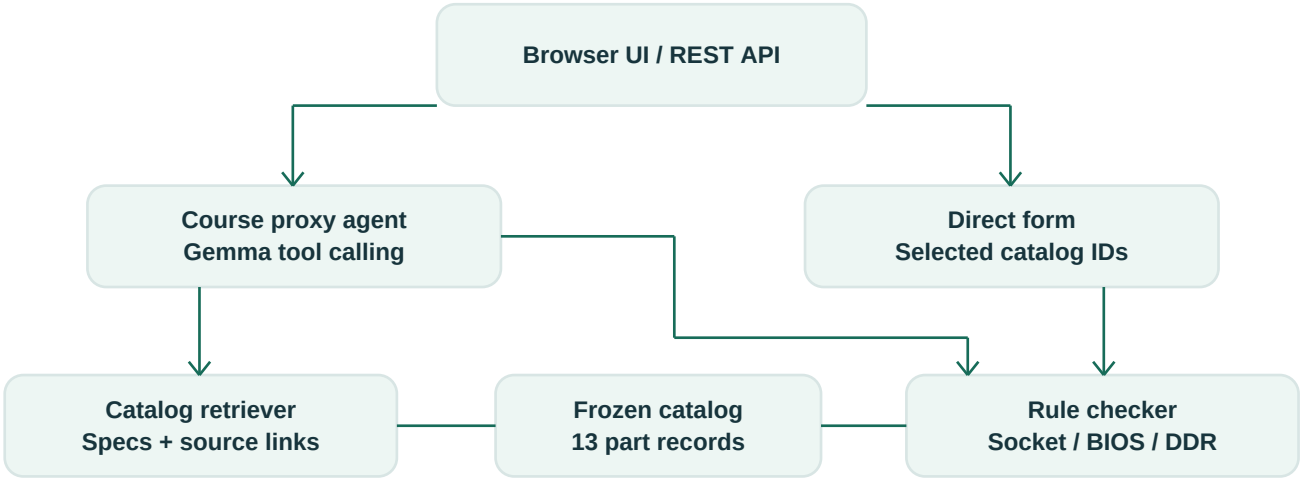

Figure 1. The browser can send a natural-language question to the agent or send selected catalog IDs directly to the checker.

Evidence retrieval

The app builds 17 documents from the catalog, with a specification per part and a separate CPU support document per board. The keyword baseline uses BM25. Hybrid search combines normalized BM25 with cosine similarity over sparse character 3- and 4-gram TF-IDF vectors. Model-name matches and spec or BIOS intent can raise relevant documents.

Tool calls and stopping

The agent sends the conversation and two tool schemas to the class proxy via the OpenAI Python client. The server validates argument keys and types. A check requires searched IDs that match the explicitly named catalog parts. The loop stops on a final response or its five-model-call limit. The UI presents the checker result separately from model text.

Verdict and configuration

The checker reports compatible, incompatible, conditional, or unknown within the rules covered. Missing evidence stays unknown. A listed BIOS minimum gives a conditional verdict when the current version is absent or below the minimum. The API key and base URL are read from the environment or the ignored local .env file.

Measured results

A frozen catalog and checked-in gold labels make the offline comparison reproducible.

Test set and procedure

The 30 prompts include Thai and English questions, exact and abbreviated model names, similar variants, conflicting hardware, and BIOS thresholds. Each prompt identifies the relevant source documents. Nineteen prompts also specify a gold verdict and part IDs. Both retrieval methods rank the same documents from the same prompt and use the same top-three cutoff.

Metrics

Recall@3 averages the fraction of relevant documents found in the top three. MRR@3 averages the inverse rank of the first relevant document, using zero if all three miss. Direct verdict accuracy is the fraction of correct verdicts given the gold part IDs.

Retriever	Prompts	Recall@3	MRR@3
BM25 baseline	30	0.9111	0.8944
Hybrid	30	0.9833	0.9778

Hybrid retrieval improved Recall@3 by 7.22 percentage points. For case 13, the compact identifier **tufb550pluswifiii** put the correct specs in first place with hybrid search. BM25 missed that document within the first three.

Direct checker and live evaluation

The direct checker matched 19 of 19 gold verdicts using canonical part IDs. This measures the rule logic on the labelled cases. Live proxy performance remains unmeasured in this report. The live evaluation command scores the agent's structured verdict once the course credentials are configured.

Reproduce offline python -m eval.run --output eval/results.json

Measure the live agent python -m eval.run --live --output eval/live-results.json

Failures and limits

The remaining retrieval miss shows where shorthand still breaks the evidence ranking.

Observed failure: comparing two B550 variants

Case 27 asks about BIOS for the plain B550-PLUS and refers to the second model only as WIFI II. Hybrid search returns the plain board's CPU support document, its specs, and the Ryzen 5 5600 specification. The WIFI II support document falls outside the top three. BM25 retrieved both support documents in this case.

The hybrid boosts explicit model mentions, so the fully spelled-out model and CPU crowd out the variant referenced only by suffix. A future comparison mode could resolve the two board names separately and retrieve one support document per board. A held-out test set of paired-model comparisons would measure whether this improves evidence coverage.

Catalog and evaluation limits

The catalog is a small snapshot. Where ASUS publishes conflicting entries for the same CPU and board, the pairing is left unresolved in the catalog. A DDR generation match does not establish kit qualification, attainable speed, or overall build compatibility. The evaluation prompts were designed for this catalog and are too few to estimate performance across vendors. The 19 direct verdicts do not include model interpretation.

Manufacturer evidence and reproducibility

ASUS TUF Gaming B650-PLUS WIFI Specifications CPU support

ASUS TUF Gaming B650-PLUS Specifications CPU support

ASUS TUF Gaming B550-PLUS WIFI II Specifications CPU support

ASUS TUF Gaming B550-PLUS Specifications CPU support

The checked-in catalog includes the CPU and RAM sources. The gold labels and per-case rankings are in eval/gold.json and eval/results.json.